Pontifícia Universidade Católica de Minas Gerais
Engenharia de Software – Noite
Banco de Dados
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Discente: Arthur Araujo Mendonça; Bruno Vieira Cardoso; Lucas Santos Rosa; Marco Willy Azevedo Gomes; Marlene Vasconcelos Moraes De Oliveira.

Lista 08:
SQL - DML

1. Criação do banco de dados ESCOLA:

```
CREATE DATABASE ESCOLA;

USE ESCOLA;
CREATE TABLE Departamento
(CodDepto INT,
 NomeDepto VARCHAR(30),
 CONSTRAINT PK_Departamento PRIMARY KEY (CodDepto));

CREATE TABLE Aluno
(Matricula INT,
 Nome VARCHAR(10),
 CodDepto INT,
 CONSTRAINT PK_Aluno PRIMARY KEY (Matricula),
 CONSTRAINT FK_AlunoDepartamento FOREIGN KEY (CodDepto) REFERENCES
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:33:31	CREATE DATABASE ESCOLA	1 row(s) affected	0.015 sec

Execução do script – criação de tabelas:

```
 CONSTRAINT PK_Historico PRIMARY KEY (Matricula,CodTurma),
 CONSTRAINT FK_HistoricoAluno FOREIGN KEY (Matricula) REFERENCES Aluno(Matricula),
 CONSTRAINT FK_HistoricoTurma FOREIGN KEY (CodTurma) REFERENCES Turma(CodTurma));

CREATE TABLE Pre_Requisito
(CodDisc VARCHAR(7),
 CodDiscPre VARCHAR(7),
 CONSTRAINT PK_Pre_Requisito PRIMARY KEY (CodDisc,CodDiscPre),
 CONSTRAINT FK_Pre_ReqDisciplina FOREIGN KEY (CodDisc) REFERENCES
Disciplina(CodDisc),
 CONSTRAINT FK_Pre_ReqDisciplinaPre FOREIGN KEY (CodDiscPre) REFERENCES
Disciplina(CodDisc));

INSERT INTO Departamento VALUES (01,'Ciencia da Computacao');
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:33:31	CREATE DATABASE ESCOLA	1 row(s) affected	0.015 sec
2	15:36:10	USE ESCOLA	0 row(s) affected	0.000 sec
3	15:36:10	CREATE TABLE Departamento (CodDepto INT, NomeDepto VARCHAR(30), CONSTRAINT PK_Departamento PRIMARY KEY (CodDepto))	0 row(s) affected	0.047 sec
4	15:37:09	CREATE TABLE Aluno (Matricula INT, Nome VARCHAR(10), CodDepto INT, CONSTRAINT PK_Aluno PRIMARY KEY (Matricula), CONSTRAINT FK_...	0 row(s) affected	0.047 sec
5	15:37:21	CREATE TABLE Disciplina (CodDisc VARCHAR(7), NomeDisc VARCHAR(40), Creditos INT, CodDepto INT, CONSTRAINT PK_Disciplina PRIMARY K...	0 row(s) affected	0.062 sec
6	15:37:28	CREATE TABLE Turma (CodTurma INT, CodDisc VARCHAR(7), Semestre VARCHAR(10), Ano INT, Professor VARCHAR(10), CONSTRAINT PK_Tur...	0 row(s) affected	0.047 sec
7	15:37:33	CREATE TABLE Historico (Matricula INT, CodTurma INT, Conceito VARCHAR(1), CONSTRAINT PK_Historico PRIMARY KEY (Matricula,CodTurma), ...	0 row(s) affected	0.062 sec
8	15:37:37	CREATE TABLE Pre_Requisito (CodDisc VARCHAR(7), CodDiscPre VARCHAR(7), CONSTRAINT PK_Pre_Requisito PRIMARY KEY (CodDisc,CodDisc...	0 row(s) affected	0.031 sec

Execução do script – inserção de tuplas das tabelas:

```
INSERT INTO Turma VALUES (119,'CC1310','Segundo',2019,'Anderson');
INSERT INTO Turma VALUES (135,'CC3380','Segundo',2019,'Samuel');
INSERT INTO Turma VALUES (100,'MAT4321','Segundo',2019,'Jane');
INSERT INTO Historico VALUES (17,112,'B');
INSERT INTO Historico VALUES (17,119,'C');
INSERT INTO Historico VALUES (8,85,'A');
INSERT INTO Historico VALUES (8,92,'A');
INSERT INTO Historico VALUES (8,102,'B');
INSERT INTO Historico VALUES (8,135,'A');
INSERT INTO Historico VALUES (10,135,'A');
INSERT INTO Historico VALUES (10,100,'A');
INSERT INTO Pre_Requisito VALUES ('CC3380','CC3320');
INSERT INTO Pre_Requisito VALUES ('CC3380','MAT2410');
INSERT INTO Pre_Requisito VALUES ('CC3320','CC1310');
```

Output

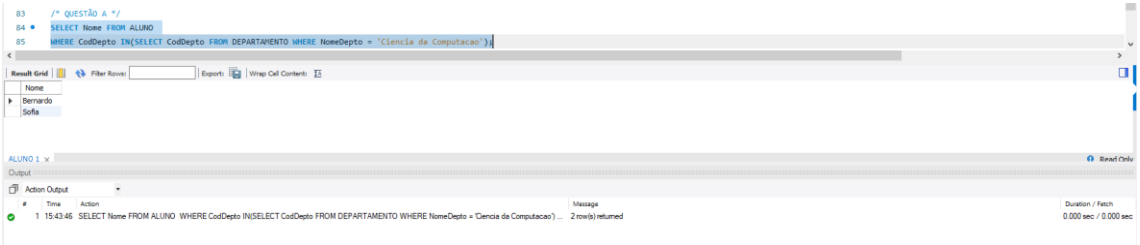
Action Output

#	Time	Action	Message	Duration / Fetch
10	15:38:38	INSERT INTO Departamento VALUES (02,'Matematica')	1 row(s) affected	0.000 sec
11	15:38:38	INSERT INTO Aluno VALUES (17,'Sofia',01)	1 row(s) affected	0.000 sec
12	15:38:38	INSERT INTO Aluno VALUES (8,'Bernardo',01)	1 row(s) affected	0.000 sec
13	15:38:38	INSERT INTO Aluno VALUES (10,'Maria',02)	1 row(s) affected	0.000 sec
14	15:38:38	INSERT INTO Disciplina VALUES (CC1310,'Introducao a Ciencia da Computacao',4,01)	1 row(s) affected	0.016 sec
15	15:38:38	INSERT INTO Disciplina VALUES (CC3320,'Estruturas de Dados',4,01)	1 row(s) affected	0.000 sec
16	15:38:38	INSERT INTO Disciplina VALUES (MAT2410,'Matematica Discreta',3,02)	1 row(s) affected	0.000 sec
17	15:38:38	INSERT INTO Disciplina VALUES (CC3380,'Banco de Dados',3,01)	1 row(s) affected	0.000 sec
18	15:38:38	INSERT INTO Disciplina VALUES (MAT4321,'Calculo I',4,02)	1 row(s) affected	0.000 sec
19	15:38:38	INSERT INTO Turma VALUES (85,MAT2410,'Segundo',2018,'Carlos')	1 row(s) affected	0.000 sec
20	15:38:38	INSERT INTO Turma VALUES (92,CC1310,'Segundo',2018,'Anderson')	1 row(s) affected	0.000 sec
21	15:38:38	INSERT INTO Turma VALUES (102,CC3320,'Primeiro',2019,'Claudio')	1 row(s) affected	0.000 sec
22	15:38:38	INSERT INTO Turma VALUES (112,'MAT2410','Segundo',2019,'Camila')	1 row(s) affected	0.000 sec
23	15:38:38	INSERT INTO Turma VALUES (119,CC1310,'Segundo',2019,'Anderson')	1 row(s) affected	0.000 sec
24	15:38:38	INSERT INTO Turma VALUES (135,'CC3380','Segundo',2019,'Samuel')	1 row(s) affected	0.000 sec
25	15:38:38	INSERT INTO Turma VALUES (100,'MAT4321','Segundo',2019,'Jane')	1 row(s) affected	0.000 sec
26	15:38:38	INSERT INTO Historico VALUES (17,112,'B')	1 row(s) affected	0.000 sec
27	15:38:38	INSERT INTO Historico VALUES (17,119,'C')	1 row(s) affected	0.000 sec
28	15:38:38	INSERT INTO Historico VALUES (8,85,'A')	1 row(s) affected	0.015 sec
29	15:38:38	INSERT INTO Historico VALUES (8,92,'A')	1 row(s) affected	0.000 sec
30	15:38:38	INSERT INTO Historico VALUES (8,102,'B')	1 row(s) affected	0.000 sec
31	15:38:38	INSERT INTO Historico VALUES (8,135,'A')	1 row(s) affected	0.000 sec
32	15:38:38	INSERT INTO Historico VALUES (10,135,'A')	1 row(s) affected	0.000 sec

2.

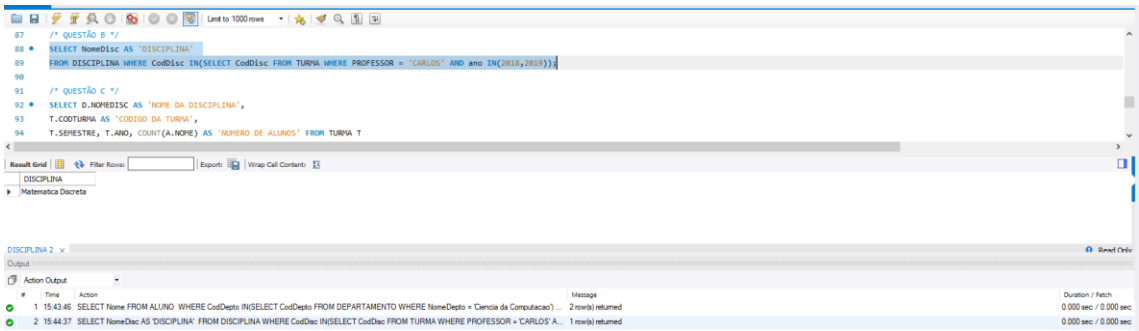
a) SELECT Nome FROM ALUNO

WHERE CodDepto IN(SELECT CodDepto FROM DEPARTAMENTO WHERE NomeDepto = 'Ciencia da Computacao');



b) SELECT NomeDisc AS 'DISCIPLINA'

FROM DISCIPLINA WHERE CodDisc IN(SELECT CodDisc FROM TURMA WHERE PROFESSOR = 'CARLOS' AND ano IN(2018,2019));



c) SELECT D.NOMEDISC AS 'NOME DA DISCIPLINA',

T.CODTURMA AS 'CODIGO DA TURMA',

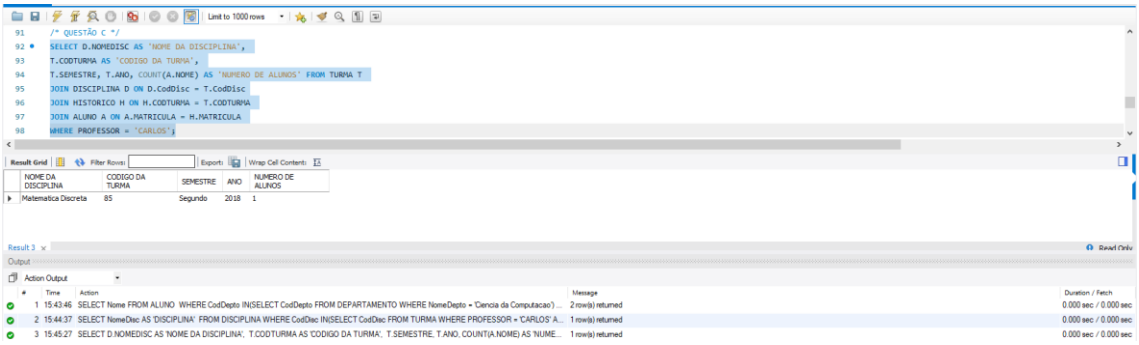
T.SEMESTRE, T.ANO, COUNT(A.NOME) AS 'NUMERO DE ALUNOS' FROM TURMA T

JOIN DISCIPLINA D ON D.CodDisc = T.CodDisc

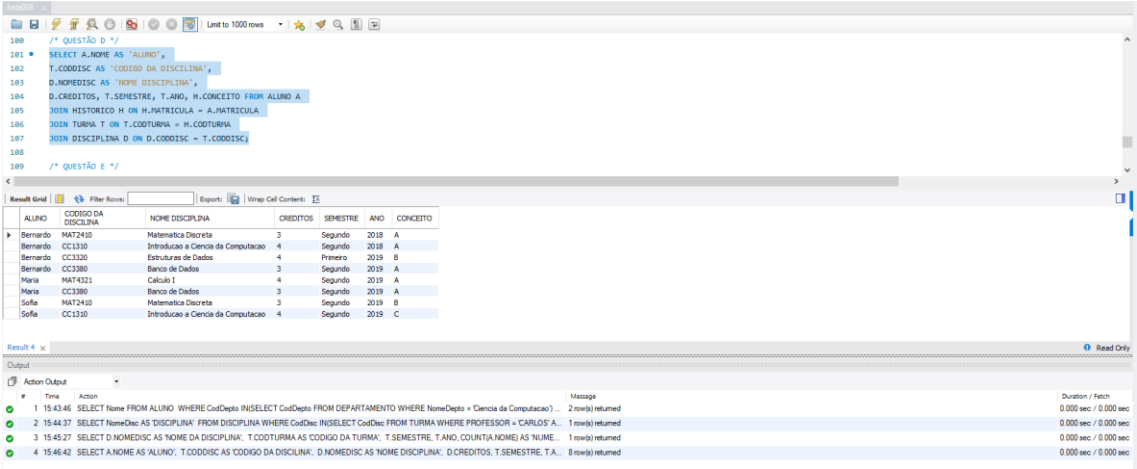
JOIN HISTORICO H ON H.CODTURMA = T.CODTURMA

JOIN ALUNO A ON A.MATRICULA = H.MATRICULA

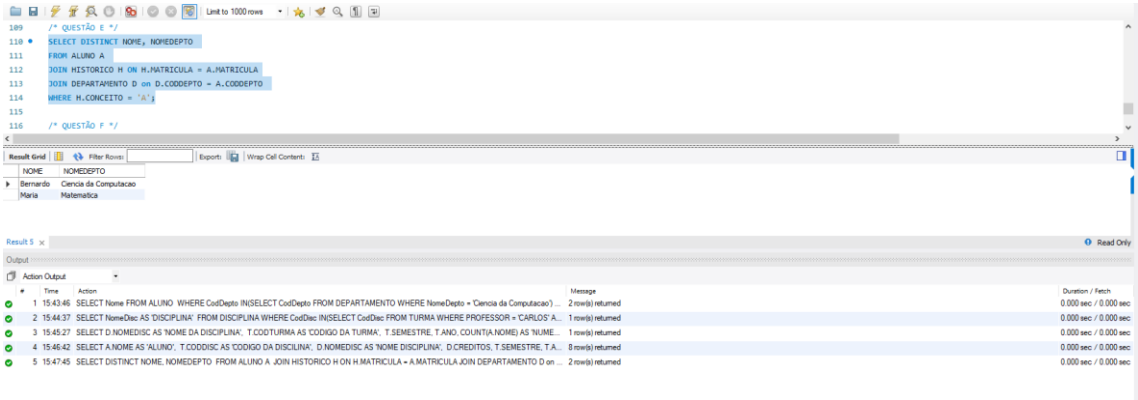
WHERE PROFESSOR = 'CARLOS';



d) SELECT A.NOME AS 'ALUNO',
T.CODDISC AS 'CODIGO DA DISCILINA',
D.NOMEDISC AS 'NOME DISCIPLINA',
D.CREDITOS, T.SEMESTRE, T.ANO, H.CONCEITO FROM ALUNO A
JOIN HISTORICO H ON H.MATRICULA = A.MATRICULA
JOIN TURMA T ON T.CODTURMA = H.CODTURMA
JOIN DISCIPLINA D ON D.CODDISC = T.CODDISC;



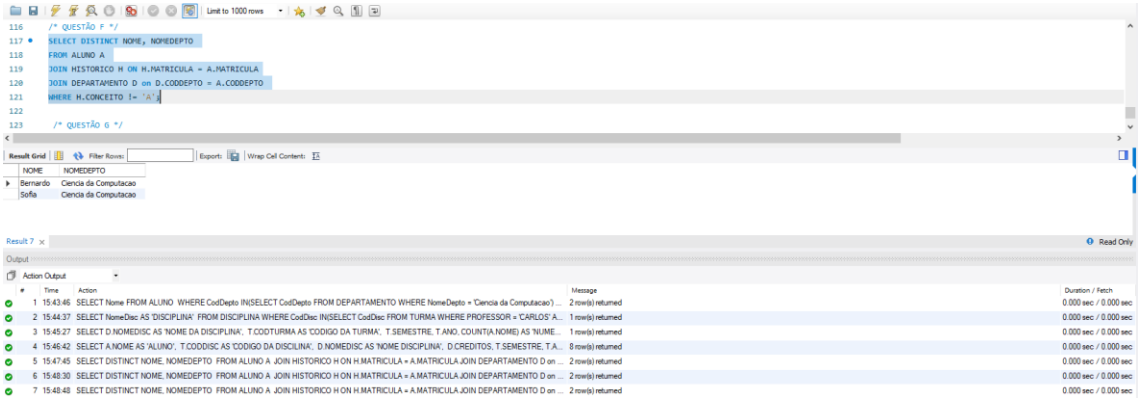
e) SELECT DISTINCT NOME, NOMEDEPTO
FROM ALUNO A
JOIN HISTORICO H ON H.MATRICULA = A.MATRICULA
JOIN DEPARTAMENTO D on D.CODDEPTO = A.CODDEPTO
WHERE H.CONCEITO = 'A';



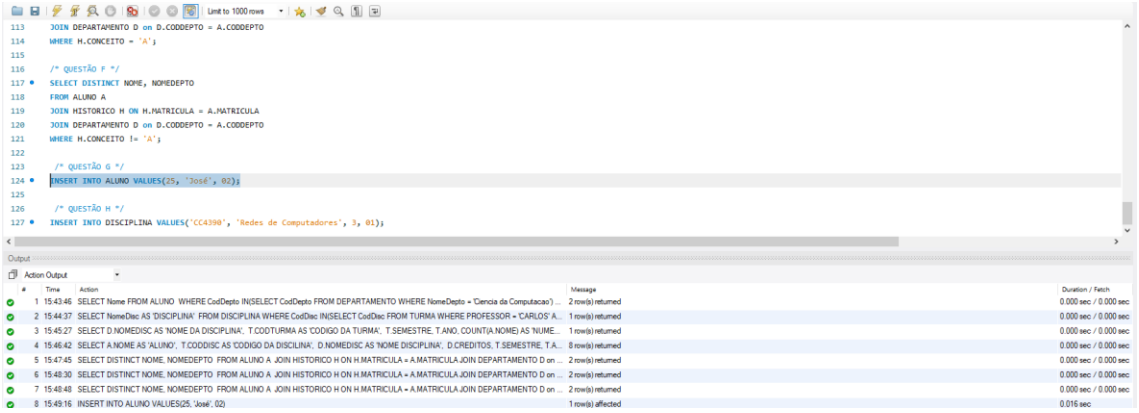
f) SELECT DISTINCT NOME, NOMEDEPTO
FROM ALUNO A
JOIN HISTORICO H ON H.MATRICULA = A.MATRICULA

JOIN DEPARTAMENTO D on D.CODDEPTO = A.CODDEPTO

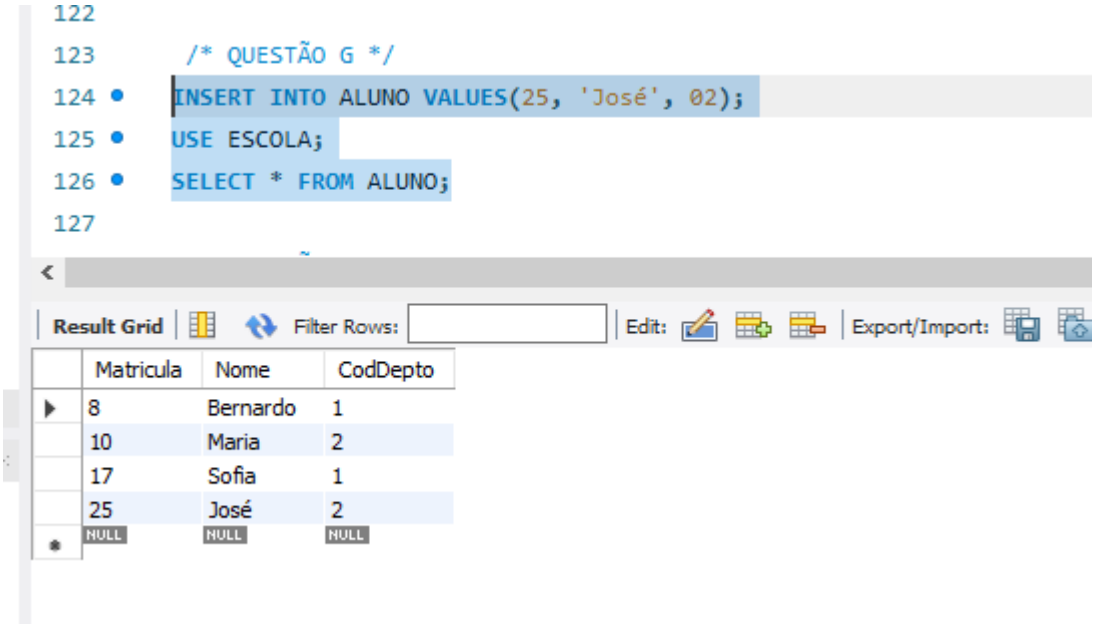
WHERE H.CONCEITO != 'A';



g) INSERT INTO ALUNO VALUES(25, 'José', 02);



Resultado da inserção:



```
/* QUESTÃO G */
INSERT INTO ALUNO VALUES(25, 'José', 02);
USE ESCOLA;
SELECT * FROM ALUNO;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Matricula	Nome	CodDepto
8	Bernardo	1
10	Maria	2
17	Sofia	1
25	José	2
NULL	NULL	NULL

h) INSERT INTO DISCIPLINA VALUES('CC4390', 'Redes de Computadores', 3, 01);

```
113     JOIN DEPARTAMENTO D on D.CODDEPTO = A.CODDEPTO
114     WHERE H.CONCEITO = 'A';
115
116     /* QUESTÃO F */
117     SELECT DISTINCT NOME, NOMEDEPTO
118     FROM ALUNO A
119     JOIN HISTORICO H ON H.MATRICULA = A.MATRICULA
120     JOIN DEPARTAMENTO D on D.CODDEPTO = A.CODDEPTO
121     WHERE H.CONCEITO != 'A';
122
123      /* QUESTÃO G */
124     INSERT INTO ALUNO VALUES(25, 'José', 02);
125
126      /* QUESTÃO H */
127     INSERT INTO DISCIPLINA VALUES('CC4390', 'Redes de Computadores', 3, 01);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:43:46	SELECT Nome FROM ALUNO WHERE CodDepto IN(SELECT CodDepto FROM DEPARTAMENTO WHERE NomeDepto = 'Ciencia da Computacao') ...	2 row(s) returned	0.000 sec / 0.000 sec
2	15:44:37	SELECT NomeDisc AS 'DISCIPLINA' FROM DISCIPLINA WHERE CodDisc IN(SELECT CodDisc FROM TURMA WHERE PROFESSOR = 'CARLOS' A...	1 row(s) returned	0.000 sec / 0.000 sec
3	15:45:27	SELECT D.NOMEDISC AS 'NOME DA DISCIPLINA', T.CODTURMA AS 'CODIGO DA TURMA', T.SEMESTRE, T.ANO, COUNT(A.NOME) AS 'NUME...	1 row(s) returned	0.000 sec / 0.000 sec
4	15:46:42	SELECT A.NOME AS 'ALUNO', T.CODDISC AS 'CODIGO DA DISCILINA', D.NOMEDISC AS 'NOME DISCIPLINA', D.CREDITOS, T.SEMESTRE, T.A...	8 row(s) returned	0.000 sec / 0.000 sec
5	15:47:45	SELECT DISTINCT NOME, NOMEDEPTO FROM ALUNO A JOIN HISTORICO H ON H.MATRICULA = A.MATRICULA JOIN DEPARTAMENTO D on ...	2 row(s) returned	0.000 sec / 0.000 sec
6	15:48:30	SELECT DISTINCT NOME, NOMEDEPTO FROM ALUNO A JOIN HISTORICO H ON H.MATRICULA = A.MATRICULA JOIN DEPARTAMENTO D on ...	2 row(s) returned	0.000 sec / 0.000 sec
7	15:48:48	SELECT DISTINCT NOME, NOMEDEPTO FROM ALUNO A JOIN HISTORICO H ON H.MATRICULA = A.MATRICULA JOIN DEPARTAMENTO D on ...	2 row(s) returned	0.000 sec / 0.000 sec
8	15:49:16	INSERT INTO ALUNO VALUES(25, 'José', 02)	1 row(s) affected	0.016 sec
9	15:49:58	INSERT INTO DISCIPLINA VALUES('CC4390', 'Redes de Computadores', 3, 01)	1 row(s) affected	0.000 sec

Resultado da inserção:

```
125
126      /* QUESTÃO H */
127 •   INSERT INTO DISCIPLINA VALUES('CC4390', 'Redes de Computadores', 3, 01);
128 •   USE ESCOLA;
129 •   SELECT * FROM DISCIPLINA;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Cont

CodDisc	NomeDisc	Creditos	CodDepto
CC1310	Introducao a Ciencia da Computacao	4	1
CC3320	Estruturas de Dados	4	1
CC3380	Banco de Dados	3	1
CC4390	Redes de Computadores	3	1
MAT2410	Matematica Discreta	3	2
MAT4321	Calculo I	4	2
NULL	NULL	NULL	NULL